# TINKERBADGE / ELECTRICAL OVERVIEW

Prototype schematic 0.1 | 2026-11-09
Source-controlled circuit contracts: docs/design/
Electrical source review precedes each populated block.

01 POWER + CHARGING

USB / dock input selection
Power path, protected BAT sensing
Single nominal 3.3 V shared rail

File: power.kicad_sch

02 PROCESSOR + RECOVERY

ESP32-S3-WROOM-1-N16R8
Native USB, UART and boot/reset
Reviewed GPIO allocation

File: mcu.kicad_sch

03 E-PAPER DISPLAY

ED037TC1 electrical interface
Switched logic and isolated inputs
Exact panel boost support

File: display.kicad_sch

04 LIGHTING + BUTTONS

Three role + five affinity pixels
Red / green power indicator
Two buttons, RTC wake

File: lighting.kicad_sch

05 SENSORS + EXPANSION

Ambient, motion, dynamic NFC
IR transmit / receive
Qwiic / SAO and status expander

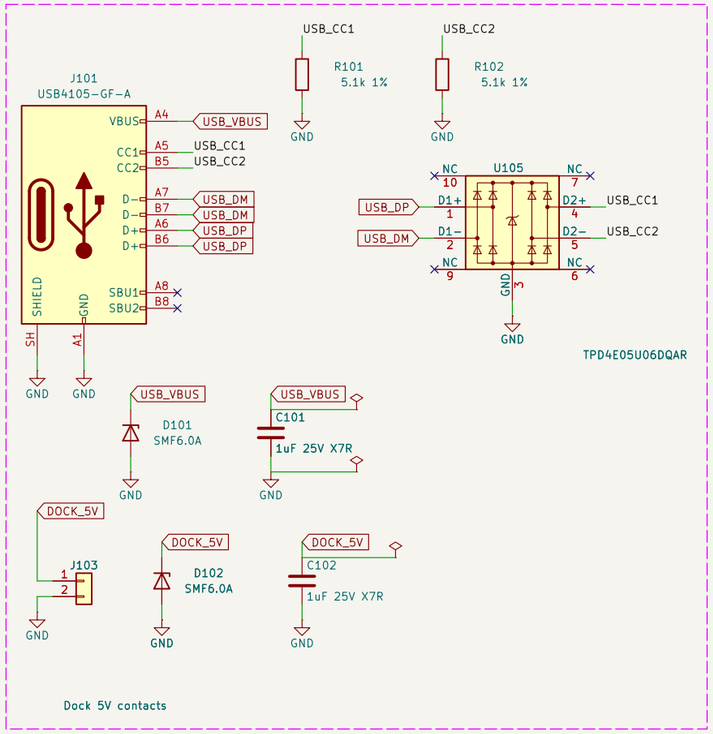
File: peripherals.kicad_sch

INITIAL VARIANT
Onboard charging is disabled.
Two-wire protected LiPo; optional NTC not fitted.
See docs/design/implementation_status.md
for physical and prototype release gates.

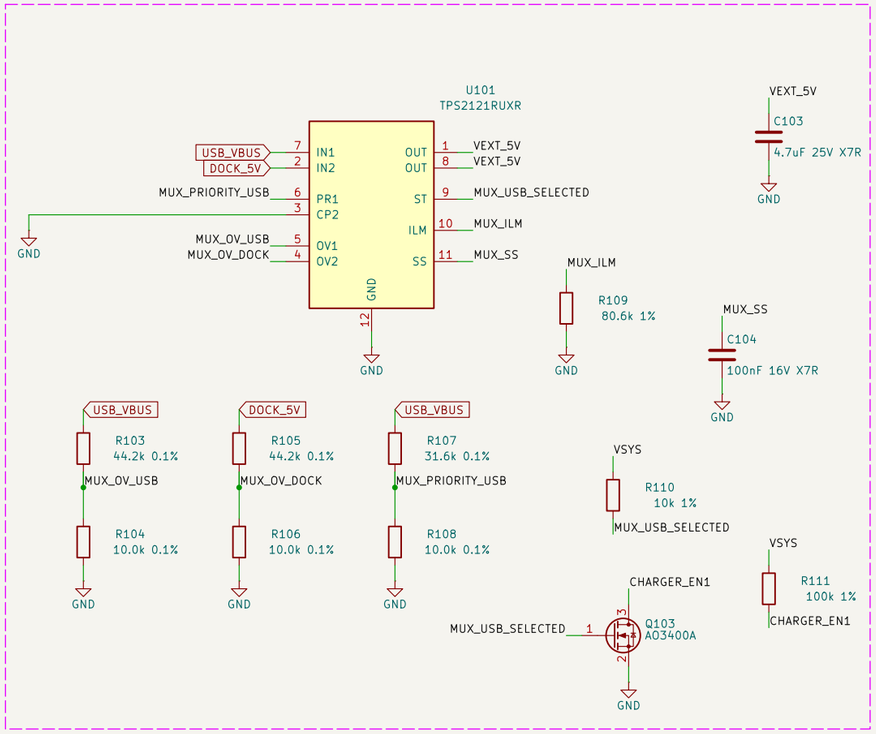
PROTOTYPE RAIL DECISION
Shared 3.3 V supply, including the eight RGB LEDs.
Tolerance can fall below the LED specified minimum.
Startup / low-battery / Wi-Fi / LED-step tests required.

| | | |
|---|---|---|
| Sheet: / | | |
| File: tinkerbadge.kicad_sch | | |
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| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.8 | | Id: 1/6 |

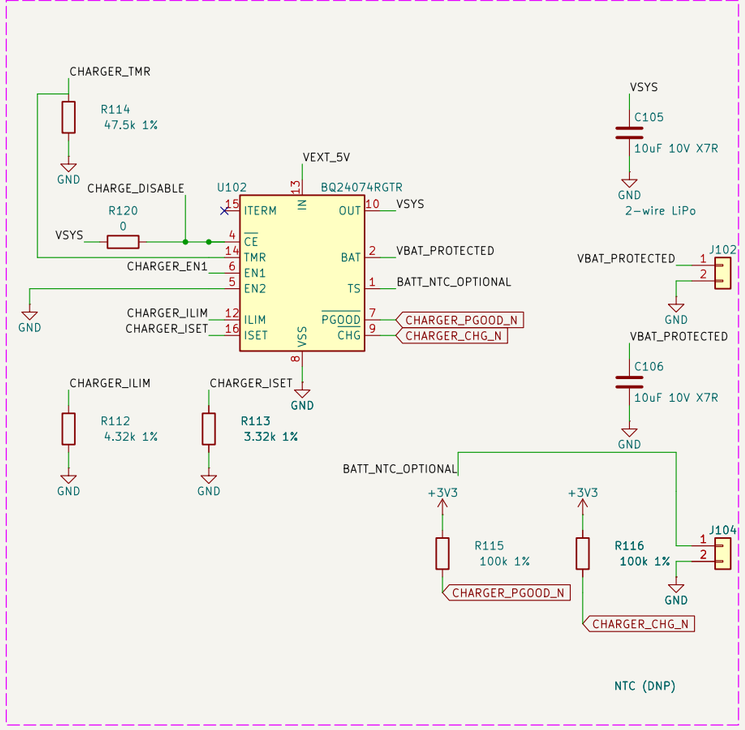
## USB / dock inputs



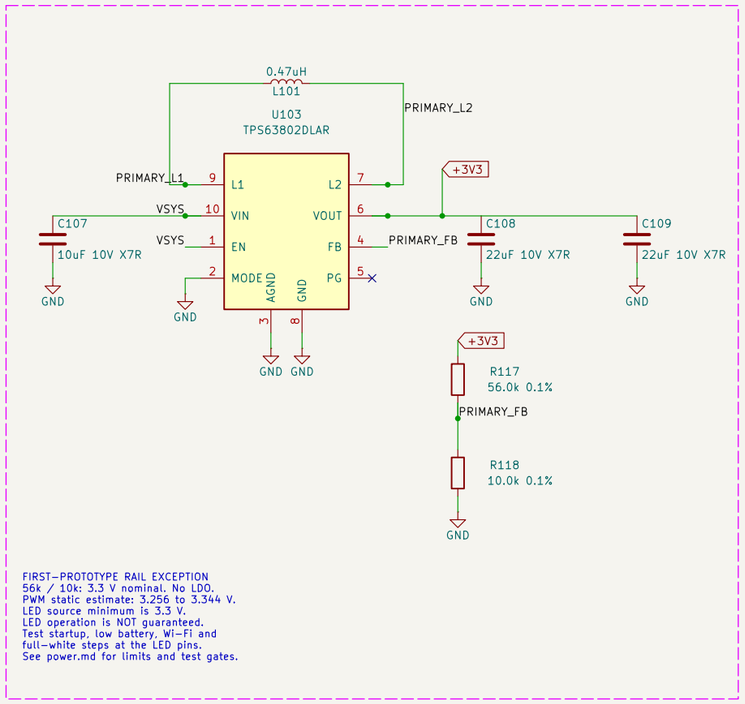

## USB-priority source selection



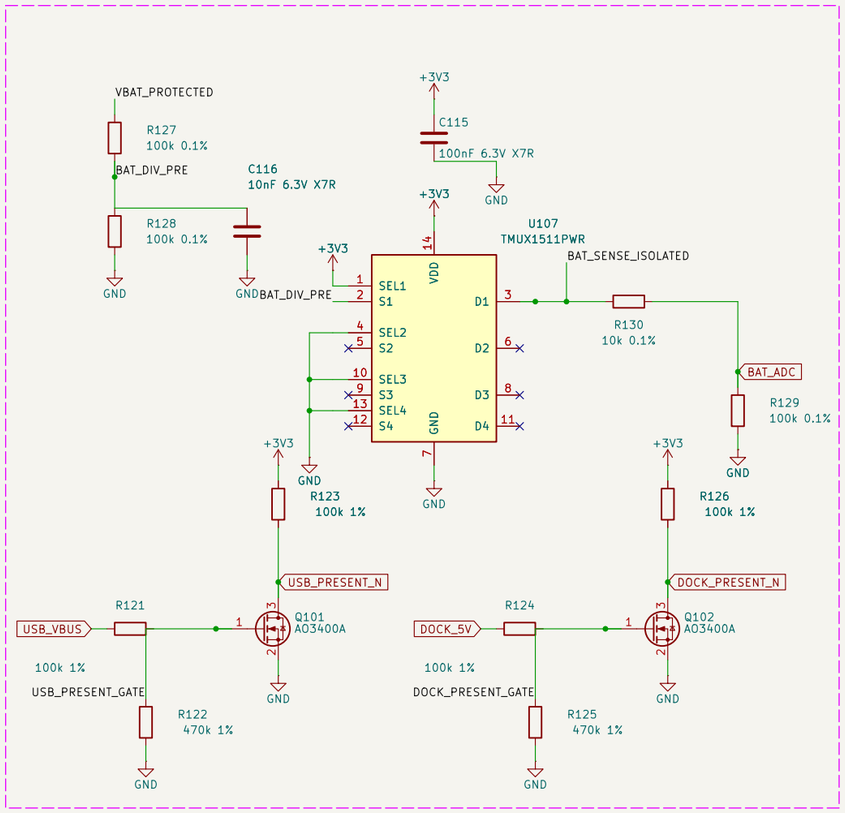

## Power path: charging disabled



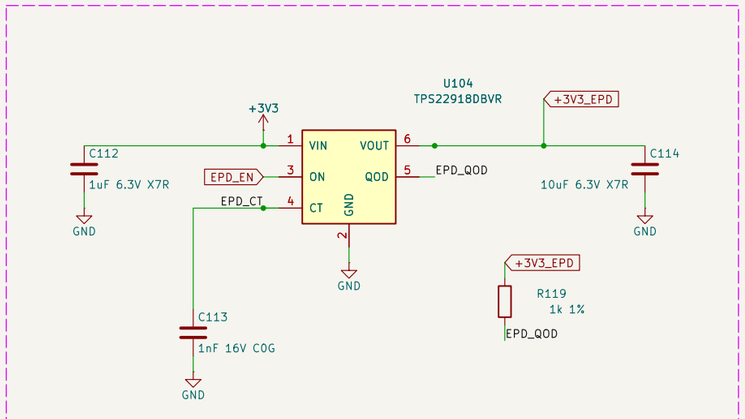

## Shared nominal 3.3 V rail




## Battery ADC / source presence




## Display power



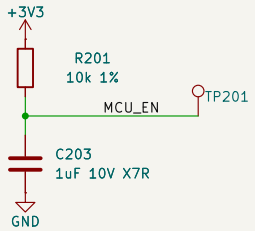

DEVELOPMENT SAFETY GATES
R120 fitted: charging OFF. J104 thermistor DNP.
Fitting a sensor alone does not enable charging.
J102 terminal numbers are logical, not vendor polarity.
Footprints / exact pack / thermal and peak loads: pending.
USB 100 mA and dock 500 mA modes need bench validation.

Electrical development draft. No PCB layout or manufacturing release. See docs/design/power.md.

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| Sheet: /01 POWER + CHARGING/ | |
| File: power.kicad_sch | |
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| KiCad E.D.A. 10.0.8 | **Rev:** |
| | Id: 2/6 |

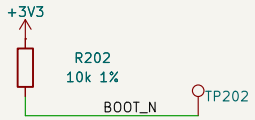
# 02 PROCESSOR + RECOVERY

## Enable + Reset




EN: 10 k / 1 uF. Pull low to reset.

## Boot / Download

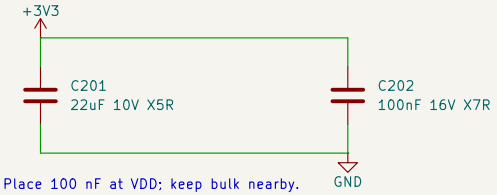



Hold BOOT low while resetting EN.

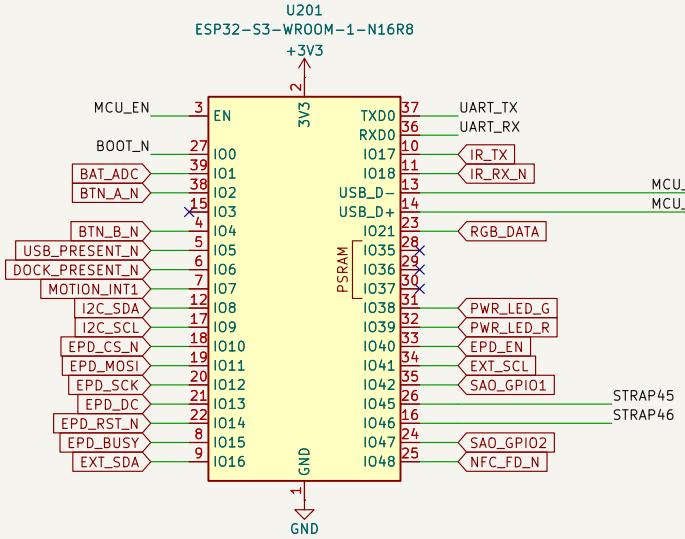
## Power Test



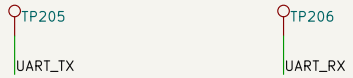

## ESP32-S3 Processor + Interfaces



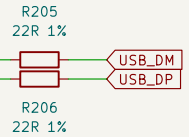

Place 100 nF at VDD; keep bulk nearby.




## UART RECOVERY




## USB 2.0 FS

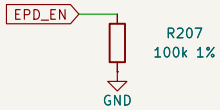



## BOOT STRAPS




Default: SPI boot, 3.3 V flash.

## DISPLAY DEFAULT OFF



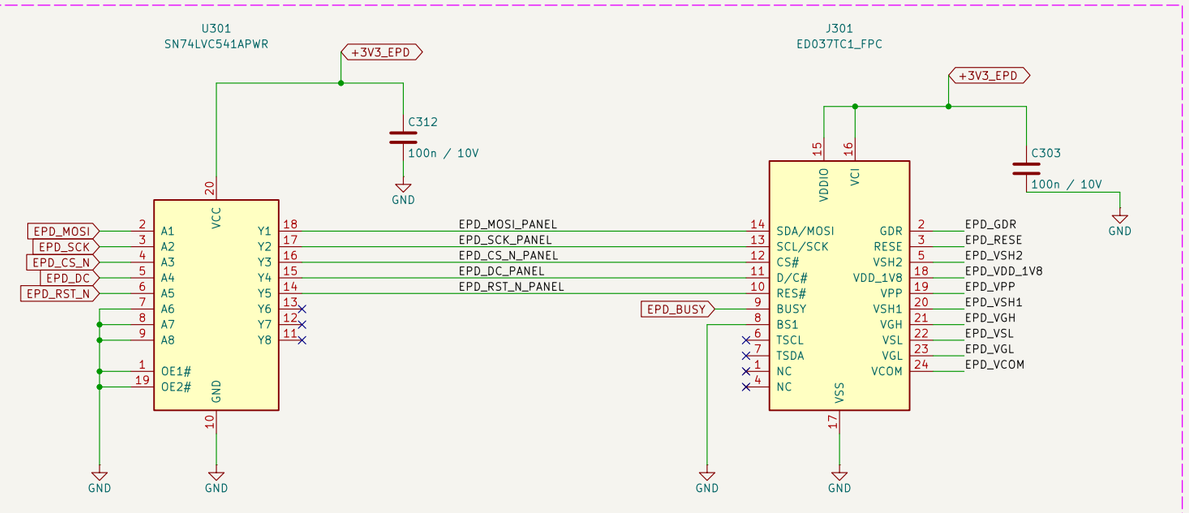

GPIO40 may go high in USB-OTG ROM recovery.
The panel remains reset by its local pull-down.

N16R8: GPIO35 / 36 / 37 reserved for octal PSRAM.
GPIO3 left open with default eFuse JTAG selection.
GND pins 1, 40 and exposed pad 41 are one net.
Antenna keepout and RF layout are PCB release gates.

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|---|---|---|
| Sheet: /02 PROCESSOR + RECOVERY/ | | |
| File: mcu.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.8 | | Id: 3/6 |

Prototype electrical capture. Physical FPC mating and panel revision must be verified before layout.

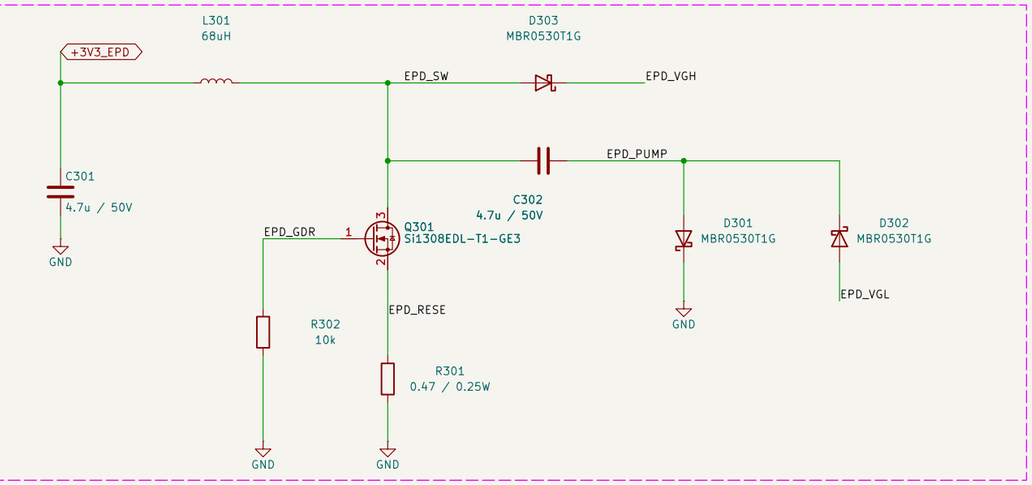
## Panel interface / 3.7-inch display



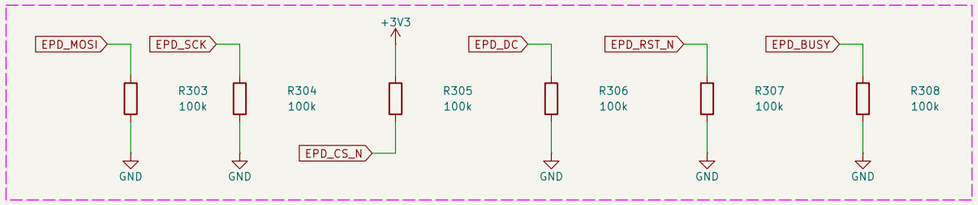

U301 isolates MCU outputs when the panel supply is off (Ioff-rated). BUSY is a pulled-down input.

Power-on: hold reset low, enable +3V3_EPD, initialize, then update. Quiesce SPI and sleep the panel before power-off.

## Panel boost and charge pump

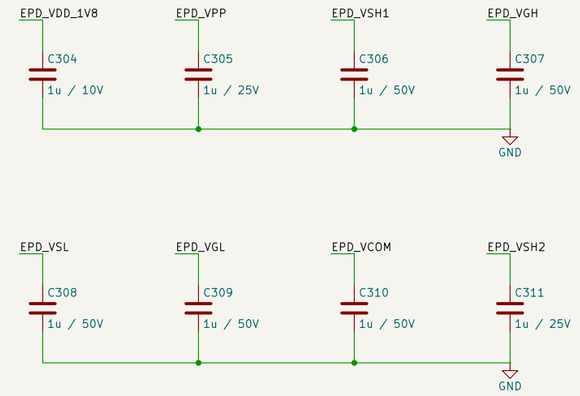



## Boot / off-state bias




## Panel rail decoupling

Use nonpolar X7R capacitors; verify capacitance under DC bias.




FPC pins 1 and 4 are NC per the raw-panel document.

Pin 5 is VSH2. VDD on pin 18 is local 1.8 V, not the 3.3 V input.

Release gates: exact panel, FPC orientation, inductor saturation, capacitor derating, measured startup/refresh/off-state waveforms.

J301 intentionally has no PCB footprint pending the mating audit.

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| Sheet: /03 E-PAPER DISPLAY/ | |
| File: display.kicad_sch | |
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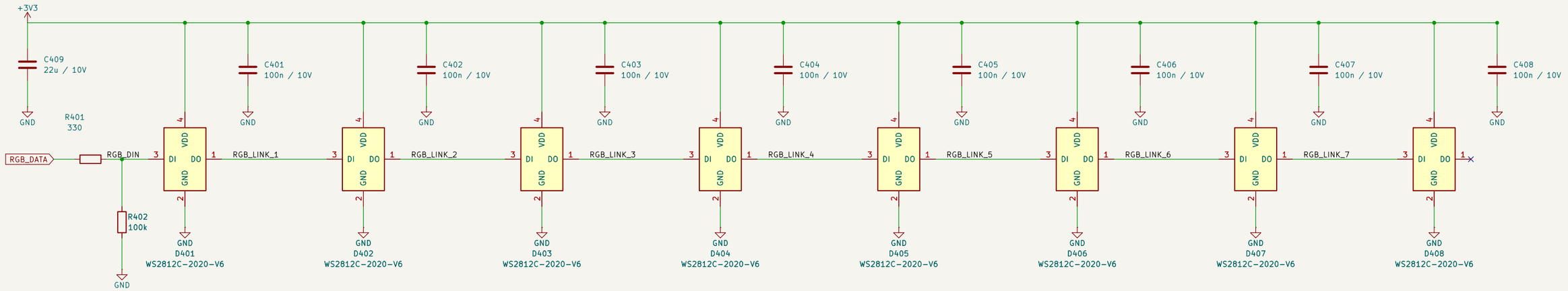
# 04 LIGHTING + BUTTONS

## Addressable RGB Chain

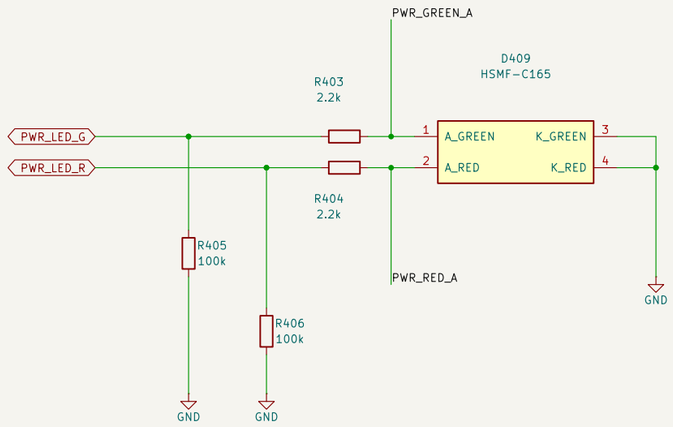
RGB chain: exact WS2812C-2020-V6 on shared nominal +3V3. Below 3.3 V operation is not guaranteed.

Send black before storage. Respect the system power/current budget in firmware.

Prototype gate: scope startup, low battery, Wi-Fi and full-white transitions at the LEDs; verify shared peak capacity.



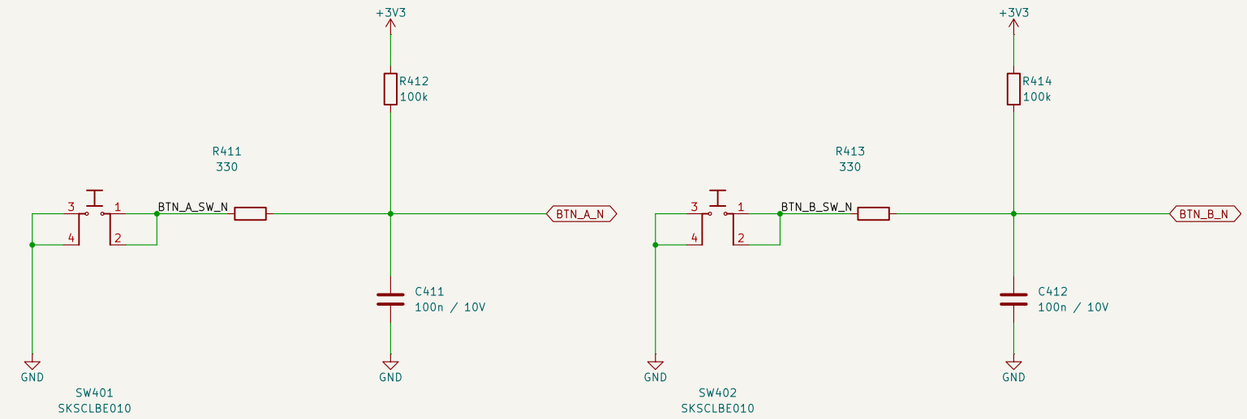

## Independent Green / Red Power Indicator




Both LED control outputs default low. Current is limited by 2.2k per die.

## User Buttons

A is the retained storage-wake button; B is a normal input.




Alps side push, 2.2 N. Common pairs: 1-2 and 3-4.

Always-on bias and capacitive loading need application qualification.

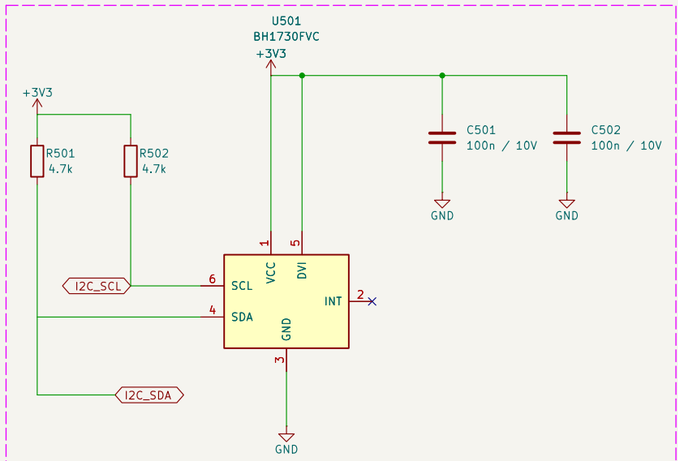
100k / 100nF gives about 10 ms debounce; firmware must also debounce.

Footprints for the exact LEDs and switch are intentionally blank pending their physical pad audit.

| | | |
|---|---|---|
| Sheet: /04 LIGHTING + BUTTONS/ | | |
| File: lighting.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.8 | | Id: 5/6 |

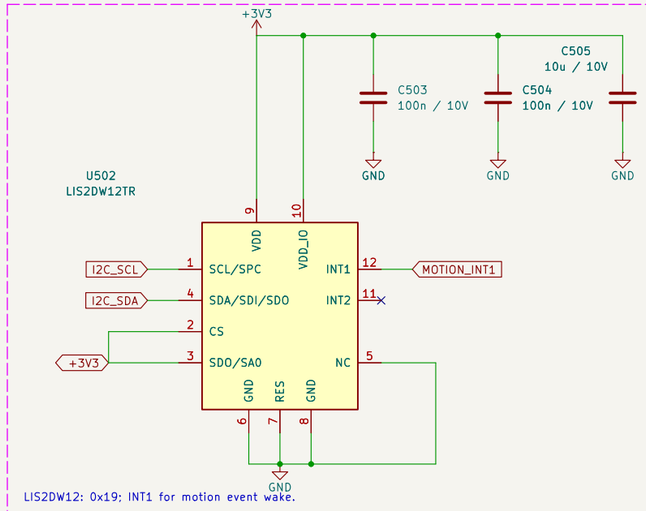
# 05 SENSORS + EXPANSION

## Ambient Light + I2C Pull-ups




BH1730: 0x29; poll ambient light. INT unused.

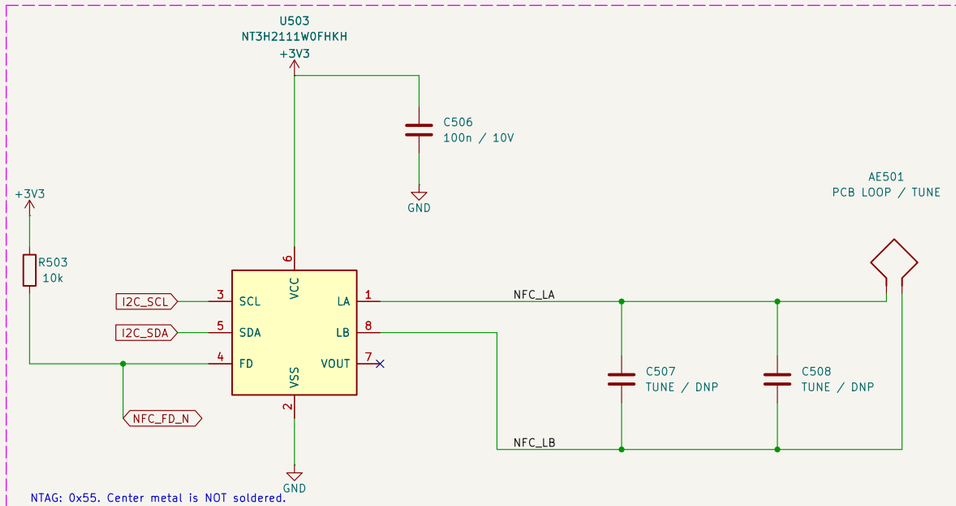
## Motion Sensor




LIS2DW12: 0x19; INT1 for motion event wake.

CS and SA0 high. Reserved pad 7 is grounded.

## Dynamic NFC Tag + Antenna




NTAG: 0x55. Center metal is NOT soldered.

Tune the PCB loop in the finished badge; tuning caps DNP.

ONBOARD I2C: 100 kHz initially. NTAG errata: STOP/START >= 50 us; guard RF edges; quiet bus for 1 ms at startup.

## Status / Enable Expander




TCA9534 address: 0x20.
Write latch 0x00 before
direction register 0xFB.

## Protected Accessory Rail + Qwiic / SAO

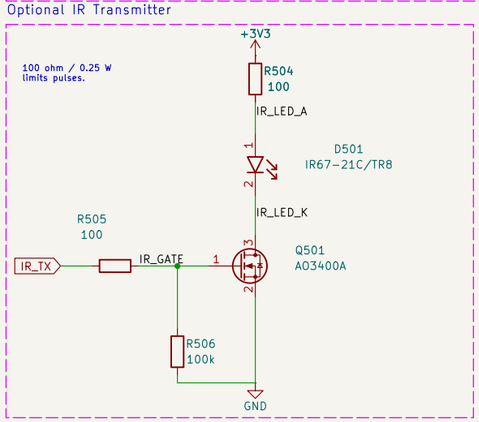
50 mA normal total. 210k / 1% sets 109–152.3 mA current limit.

Qwiic + SAO: 3.3 V only, 50 mA shared budget.

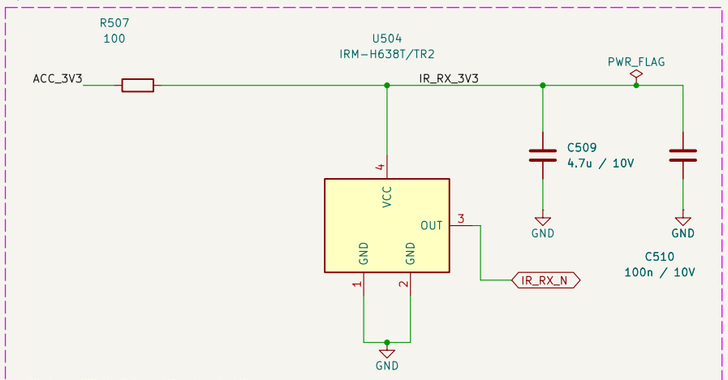
No hot-plug, external power or off-state GPIO drive.




## Optional IR Transmitter




## Optional IR Receiver




IR RX shares ACC_EN. Quiesce I/O before disabling.

Sheet: /05 SENSORS + EXPANSION/
File: peripherals.kicad_sch

**Title:**

| Size: A3 | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 6/6 |